

EXAMINATION INFORMATION PAGE

Home exam

General information about the exam:

Subject code: BID3000**Subject name:** Business Intelligence & Data Warehousing**Responsible course manager:** Peyman Teymoori**Campus:**

Ringerike

Faculty: HH**Assignment given (date and time):** 03.11.2025 09:00**Submission time (date and time):** 17.11.2025 14:00**Number of assignments:** 4 Tasks (each task includes subtasks)**Number of attachments:** [Click or tap here to enter text.](#)**Number of pages (including front page and attachments):** 9

Important information for all types of home exams:

A. EXAMINATION SUPPORT MATERIAL AND COOPERATION:

Permitted examination support material: All examination support material permitted.

Examination form (tick off): Individual ☐ Group ☒

In an individual home exam, it is not allowed to collaborate with others on how to write the answer. The text, method for solving the task, academic reflections, etc. (your answer) that is submitted for the exam should be your own work, or the group's if it is a group exam. If it is allowed to use all support materials for the exam, and it is not stated otherwise in the exam paper, you can use AI when answering the assignment. If you use AI tools to generate text, images, code, calculations, suggestions for solutions or similar, you must refer to the tool you have used. You will also need to explain, and sometimes document, how you used the AI tool.

Here you can read more about [USN's guidelines for the use of AI](#).

We would like to remind you that plagiarism control will be conducted on all answers, and unnatural similarities, illegal use of aids, AI etc. - or if you have not written the answer yourself - will result in

USN opening a case of cheating. Cheating can result in the annulment of the exam, loss of the right to take exams at USN and other universities and university colleges, and temporary suspension from USN for up till two years. [Description of individual exams and illegal collaboration can be found on the USN intranet.](#)

B. SOURCE REFERENCE:

1. You shall refer to all sources you use throughout the text.
2. Direct quotations in the text must be marked with quotation marks, followed by source reference.
3. The literature list shall be in the end of the answer paper and shall include all sources that have been used. Remember: All sources that are given throughout the text must also be in the literature list and vice versa

Criteria for the answer paper:

Font: Times New Roman

Font size: 12

Line spacing: 1

Max number of pages (excluding first page and attachments,): 12

Exam assignment:**Exam Overview**

This exam assesses students' comprehensive understanding and practical application of data warehousing concepts, ETL process implementation, dimensional modeling, and advanced analytical capabilities through SQL query development, Python analytics, and business intelligence visualization.

Exam Format

- **Group Size:** Teams of 3-6 students
- **Required Tools:**
 - **ETL:** Pentaho Data Integration (PDI), Python, or any ETL tool of choice
 - **Database:** PostgreSQL
 - **Visualization:** PowerBI
 - **Analytics:** Python
- **Submission Requirements:**

- Complete technical report (8-12 pages) in PDF format
 - All project files packaged in a zip file including:
 - ETL transformation files or scripts
 - Database scripts (.sql)
 - Dashboard file (format depends on BI tool chosen)
 - Analysis scripts
 - Supporting documentation and screenshots
-

Dataset Selection (Choose ONE)

Students must select **ONE** of the following approved datasets for their complete project implementation:

Option A: Brazilian E-commerce Dataset (Olist)

Source: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Domain: E-commerce Analytics

Key Tables: orders, order_items, customers, products, sellers, reviews, payments, geolocation

Business Context: Comprehensive e-commerce platform data enabling analysis of customer behavior, sales performance, logistics, and market insights across Brazilian regions.

Option B: Instacart Market Basket Analysis Dataset

Source: <https://www.kaggle.com/datasets/psparks/instacart-market-basket-analysis>

Domain: Grocery Retail Analytics

Key Tables: orders, products, departments, aisles, order_products_train/prior

Business Context: Online grocery shopping data focusing on customer purchasing patterns, product relationships, and reorder behavior analysis.

Option C: MovieLens 20M Dataset

Source: <https://www.kaggle.com/datasets/grouplens/movielens-20m-dataset>

Domain: Entertainment/Media Analytics

Key Tables: ratings, movies, tags, genome-scores, genome-tags, links

Business Context: Movie recommendation system data enabling analysis of user preferences, content popularity, genre trends, and rating patterns.

Option D: Online Retail II (UCI)

Source: <https://www.kaggle.com/datasets/mashlyn/online-retail-ii-uci>

Domain: International Retail Analytics

Key Features: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country

Business Context: Transactional data from a UK-based online retailer, ideal for customer segmentation, RFM analysis, and international sales patterns.

Option E: RetailRocket Recommender System Dataset

Source: <https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset>

Domain: E-commerce Behavioral Analytics

Key Tables: events (view, addtocart, transaction), item_properties, category_tree

Business Context: User behavior data from a real-world e-commerce website, perfect for recommendation systems and customer journey analysis.

Option F: Walmart M5 Forecasting Dataset

Source: <https://www.kaggle.com/competitions/m5-forecasting-accuracy/data>

Domain: Retail Sales Forecasting

Key Features: sales_train, calendar, sell_prices, sample_submission

Business Context: Hierarchical sales data from Walmart stores, ideal for demand forecasting, inventory optimization, and seasonal analysis.

Option G: NYC Yellow Taxi Trip Data

Source: <https://www.kaggle.com/datasets/elemento/nyc-yellow-taxi-trip-data>

Domain: Transportation Analytics

Key Features: pickup/dropoff times and locations, trip distances, fare amounts, payment types, passenger counts

Business Context: Urban transportation data enabling analysis of traffic patterns, demand forecasting, and fare optimization.

Assignment Tasks

Task 1: Data Warehouse Design and ETL Implementation (30 Points)

Requirements:

A. Dimensional Model Design (12 points)

- Design a data warehouse schema using **star or snowflake schema** architecture
- Implement at least **2 fact tables** and **4 dimension tables** appropriate for your chosen dataset
- Include proper primary keys and foreign keys
- Create an Entity-Relationship Diagram (ERD) showing all relationships
- Document design rationale including grain definition for each fact table

B. ETL Process Implementation (12 points)

- Using your chosen ETL tool (Pentaho, Python, or other), create ETL workflows that:
 - Extract data from source files
 - Implement basic data quality checks and cleansing
 - Handle missing values
 - Create surrogate keys for dimension tables
 - Load data into your dimensional model
 - **Optional Bonus (+3 points):** Implement one Slowly Changing Dimension (SCD Type 1 or Type 2)

C. Documentation (6 points)

- Brief documentation of warehouse design decisions
- ETL process explanation with key screenshots

- Data quality issues identified and how they were handled

Deliverables:

- Database creation scripts
- ETL transformation files or scripts
- ERD diagram
- Technical documentation

Task 2: Advanced Analytics Implementation (30 Points)**Requirements:****A. SQL Analytical Queries (15 points)**

Develop **6 analytical SQL queries** demonstrating dimensional modeling concepts:

1. **Time-based Analysis** (1 query)
 - Trend analysis or year-over-year comparison
2. **Aggregation Operations** (1 query)
 - Multi-level aggregation with GROUP BY and ROLLUP/CUBE
3. **Window Functions** (1 query)
 - Ranking or moving average calculations
4. **Complex Filtering** (1 query)
 - Subqueries or EXISTS/IN clauses
5. **Business Metrics** (2 queries)
 - KPI calculations specific to your domain
 - Customer or product performance analysis

B. Python Analytics Integration (15 points)

- Connect to your data warehouse using your chosen analytical language
- Implement **2 analytical models**:
 - **Descriptive Analytics**: Statistical summary and visualization
 - **Predictive Analytics**: One machine learning model (regression, classification, or clustering)
- Generate insights that complement your SQL analysis

Deliverables:

- SQL script file with all queries and comments
- Analytics scripts with complete analysis
- Business interpretation of findings

Task 3: Business Intelligence Dashboard (30 Points)

Requirements:**A. Interactive Dashboard Design (20 points)**

Create a dashboard using your chosen BI tool that includes:

- **Main Dashboard Page:** Key KPIs and performance metrics
- **Detailed Analysis Page:** Drill-down capabilities for deeper insights
- **Optional Third Page:** Additional analysis or what-if scenarios

Dashboard Must Include:

- At least **6 different visualization types**
- Interactive filters for dynamic analysis
- Calculated measures using the tool's formula language
- Cross-filtering between visualizations (if supported by tool)

B. Advanced Features (10 points)

- Implementation of at least **2 advanced calculations** (time intelligence, statistical measures, etc.)
- Use of any advanced features available in your chosen tool (AI features, custom visuals, etc.)
- Professional design following best practices

Deliverables:

- Complete dashboard file
- Dashboard screenshots
- Brief user guide

Task 4: Project Report and Presentation (10 Points)**Requirements:**

- Executive summary (1 page)
- Technical implementation overview (3-4 pages)
- Key insights and findings (2-3 pages)
- Business recommendations (1-2 pages)
- Team member contributions

Simplified Technical Requirements**Database Requirements:**

- Proper dimensional model (star or snowflake schema)
- Basic indexing strategy
- Data integrity constraints

ETL Requirements:

- Complete data pipeline from source to warehouse
- Basic error handling
- Data quality checks

Analytics Requirements:

- SQL queries demonstrating analytical capabilities
- Statistical/ML analysis with business context
- Clear visualization of results

Visualization Requirements:

- Professional, user-friendly dashboard
 - Interactive elements for exploration
 - Clear presentation of KPIs
-

Evaluation Criteria**Technical Implementation (35% - 35 points)**

- **Database Design:** Proper dimensional modeling
- **ETL Process:** Functionality and data quality handling
- **Query Development:** SQL correctness and complexity
- **Analytics:** Appropriate methodology and code quality
- **Dashboard:** Design quality and interactivity

Analytical Insights (30% - 30 points)

- **Business Understanding:** Relevance of analytical findings
- **Methodology:** Correct application of techniques
- **Insight Quality:** Depth and actionability of discoveries
- **Integration:** How well different analyses complement each other

Documentation and Presentation (25% - 25 points)

- **Report Quality:** Structure, clarity, completeness
- **Technical Documentation:** Code comments, process explanations
- **Visual Design:** Dashboard aesthetics and usability
- **Team Collaboration:** Evidence of effective teamwork

Innovation and Creativity (10% - 10 points)

- **Creative Solutions:** Novel approaches to challenges

- **Tool Utilization:** Effective use of chosen technologies
 - **Business Value:** Practical applicability of insights
 - **Going Beyond Requirements:** Optional features or analyses
-

Flexibility Guidelines

Tool Selection Freedom:

- Teams may choose tools based on their expertise and dataset requirements
- Justify tool selection in the report
- Ensure all team members can work with chosen tools

Workload Management:

- Focus on quality over quantity
- Optional features are for bonus points only
- Distribute work evenly among team members

Dataset-Specific Adaptations:

- Requirements may be adapted based on dataset characteristics
 - Consult with instructor if dataset limitations affect requirements
 - Document any adaptations made
-

Grading Scale

- **A:** 89-100 points (Outstanding work)
 - **B:** 77-88 points (Very good work)
 - **C:** 65-76 points (Good work)
 - **D:** 53-64 points (Satisfactory work)
 - **E:** 40-52 points (Acceptable work)
 - **F:** 0-39 points (Insufficient work)
-

Submission Guidelines

File Organization:

TeamName_BID3000_2025.zip

|— Report/

| |— BID3000_Final_Report.pdf

|— Database/


```
| ├── schema_creation.sql
|   └── queries.sql
└── ETL/
    ├── [ETL files or scripts]
    ├── Analytics/
    │   ├── [Analysis scripts]
    │   └── Dashboard/
    │       ├── [Dashboard file]
    │       └── screenshots/
    └── Documentation/
        └── ERD.pdf
```

Submission Requirements:

1. All files properly named and organized
 2. Code must be executable with clear instructions
 3. Report must include all required sections
 4. Dashboard must be functional (or provide screenshots if tool-specific)
-

Important Notes**Team Collaboration:**

- Use version control (Git) for better collaboration
- Document individual contributions
- Regular team meetings recommended

Support and Resources:

- Office hours available for tool-specific questions
- Online tutorials and documentation encouraged
- Peer learning and knowledge sharing welcomed

Academic Integrity:

- All work must be original
- External resources must be cited
- AI assistance must be documented
- Dataset sources must be acknowledged